

Full Stack Development and Deployment of a Web-Based Speech Recognition System

Week 3
26 January – 2 February 2026

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Project Overview

- MAGOR is a web-based audio and video transcription system that converts recordings into text
- **Existing Problem:**
 - Although functional, components remain incomplete and require refinement
- **Proposed Solution:**
 - Improve robustness, usability, and AI-driven recording analysis



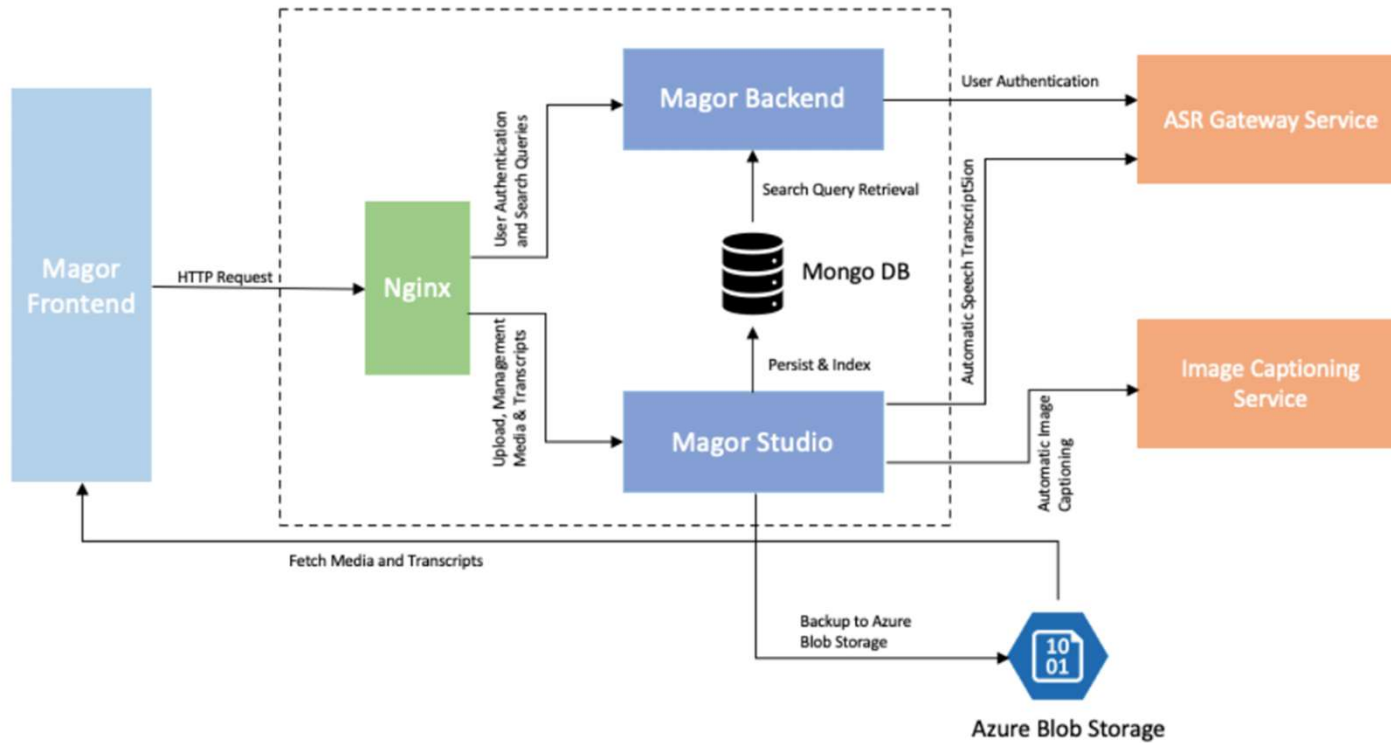
Project Objectives

The primary objectives of this project are:

- Fix system bugs and restore functionality
- Improve usability and user experience
- Enhance maintainability and scalability
- Add AI features to existing pages



Systems Overview



Source: Huang (2021)



Agenda

Work done this week:

1. Implemented transcript download on the Studio page
2. Enabled transcript downloads in multiple formats
3. Implemented media download on the Studio page
4. UI refinements



Task 1 (Transcripts download on Studio page)

- Implemented download button for each transcript in the Studio page
 - Each download corresponds to the specific transcript

Transcripts

Name:

Transcript ID: 697834accf312b002376cc74 Created At: 6 days ago

Auto Generated

Name:

Transcript ID: 6978344ccf312b002376cc55 Created At: 6 days ago

Auto Generated



Transcripts

Name:

Transcript ID: 6980f596438c1d0023c35f08 Created At: a few seconds ago

Auto Generated

Name:

Transcript ID: 6980f554438c1d0023c35ef4 Created At: a minute ago

Auto Generated



Task 1 (Transcripts download on Studio page)

- Demo:

The screenshot displays the Magor Studio web application interface. At the top, a blue header bar contains the text "Magor Studio" and a user profile icon labeled "test". Below the header, there are two buttons: "Go to Recording" and "Go to Magor Transcript Studio". The main content area is divided into two sections. On the left, there is a form for video details with fields for "Title" (containing "speech video test"), "Description" (with a placeholder "Enter description for the video"), and "Tags" (containing "video x"). A green "Save" button is located below the tags field. Below the form, there are four tabs: "Basic Details", "Transcripts", "Captions", and "Publish". The "Transcripts" tab is currently selected. On the right side of the interface, there is a video player showing a man speaking at a podium. Below the video player are three icons: a play button, a mute button, and a download button. At the bottom of the interface, there is a section titled "Auto Transcribe" with a sub-section "Current Running Jobs". A grey bar below this section contains the text "No jobs running currently.".

localhost/studio/6980f4fb438c1d0023c35ee0

Magor Studio test

Go to Recording Go to Magor Transcript Studio

Title

speech video test

Description

Enter description for the video

Tags

video x

Save

Basic Details Transcripts Captions Publish

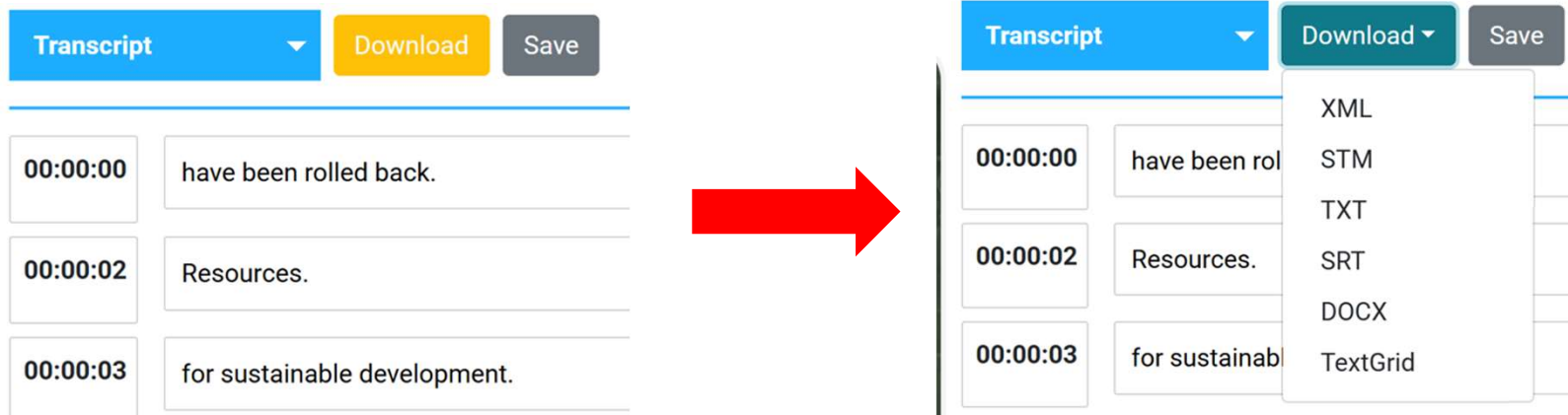
Auto Transcribe

Current Running Jobs

No jobs running currently.

Task 2 (Transcript downloads in multiple formats)

- Transcript downloads to support multiple file formats (previously only XML)
 - Formats supported:
 - XML, STM, TXT, SRT, DOCX, TEXTGRID



Task 2 (Transcript downloads in multiple formats)

- Demo:

The screenshot displays the Magor web interface. At the top, there is a blue header with the 'Magor' logo, a search bar, and a dropdown menu for 'This recording'. Below the header, the main content area features a video player on the left and a transcript on the right. The video player shows a news segment titled 'ASIA Business Tonight' with a background image of the Petronas Towers. The transcript is titled 'Indonesian stocks plunges' and contains three entries:

- 00:00:00** oil and business sharp declines for Indonesian shares, the benchmark plunging nearly 9% at one point, triggering a 30 minutes circuit breaker. The Jakarta Composite ending down more than 7% after MSCI warned about the market's investability. DNS provider wants regulators to address concerns that major companies are thinly traded and dominated by a few wealthy owners. It's warning of a possible downgrade if progress is not made by May. Indonesia's rupiah steady today. It has been under sustained pressure for months, tumbling to an all time low against the U.S. dollar.
- 00:00:39** currency has been sliding since August 2025 on concerns about fiscal discipline, or is deepening after the resignation of veteran finance Sri Mulliani Indrawati in September the same month. Uh, Bank Indonesia made two surprise rate cuts in line with the government's pro-growth push, raising concerns over its independence. Those concerns deepening in January with the nomination of President Proposal's nephew for Bank Indonesia's deputy governor with a CNA Channi to explain more about this, uh, stock market route
- 00:01:13** Uh, Shawnie.

The video player shows 'Nothing is playing right now.' and a progress bar at 0:00 / 3:50. The transcript has a 'Download' button and a 'Save' button. A 'Go to Magor Transcript Studio' link is also present.

Task 2 (Transcript downloads in multiple formats)

This XML file does not appear to have any style information associated with it. The document tree is shown below.

```
▼<AudioDoc name="5e9cdc4d-637b-445e-bf07-336b5f41c005-1770093788532">
  ▼<SegmentList>
    ▼<SpeechSegment band="S" channel="channel1" dur="37.54" gender="1" lang="eng" score="1.00" spkrid="channel1_S1" stime="0.09">
      <Word dur="0.38" score="1.0" stime="0.09">oil</Word>
      <Word dur="0.38" score="1.0" stime="0.47">and</Word>
      <Word dur="0.38" score="1.0" stime="0.84">business</Word>
      <Word dur="0.38" score="1.0" stime="1.22">sharp</Word>
      <Word dur="0.38" score="1.0" stime="1.59">declines</Word>
      <Word dur="0.38" score="1.0" stime="1.97">for</Word>
      <Word dur="0.38" score="1.0" stime="2.34">Indonesian</Word>
      <Word dur="0.38" score="1.0" stime="2.72">shares,</Word>
      <Word dur="0.38" score="1.0" stime="3.10">the</Word>
      <Word dur="0.38" score="1.0" stime="3.47">benchmark</Word>
      <Word dur="0.38" score="1.0" stime="3.85">plunging</Word>
      <Word dur="0.38" score="1.0" stime="4.22">nearly</Word>
      <Word dur="0.38" score="1.0" stime="4.60">9%</Word>
```

XML



Task 2 (Transcript downloads in multiple formats)

```
e-1770095240637 1 channel1_S1 0.09 34.63 <o,f0,male> oil and business sharp declines for Indonesian shares, the benchmark plunging nearly 9% at one point, triggering a 30 minutes circuit breaker. The Jakarta Composite ending down more than 7% after MSCI warned about the market's instability. DNS provider wants regulators to address concerns that major companies are thinly traded and dominated by a few wealthy owners. It's warning of a possible downgrade if progress is not made by May. Indonesia's rupiah steady today. It has been under sustained pressure for months, tumbling to an all time low against the U.S. dollar.
e-1770095240637 1 channel1_S1 39.15 70.46 <o,f0,male> currency has been sliding since August 2025 on concerns about fiscal discipline, or is deepening after the resignation of veteran finance Sni Mulliani Indrawati in September the same month. Uh, Bank Indonesia made two surprise rate cuts in line with the government's pro-growth push, raising concerns over its independence. Those concerns deepening in January with the nomination of President Proposal's nephew for Bank Indonesia's deputy governor with a CNA Channi to explain more about this, uh, stock market route
e-1770095240637 1 channel1_S1 73.15 74.27 <o,f0,male> Uh, Shawnie.
e-1770095240637 1 channel1_S1 76.65 88.63 <o,f0,male> concerns, confidence, both about central bank and about fiscal disciplines. In Indonesia, what will this mean this route for confidence? That's already at this point fragile.
e-1770095240637 1 channel1_S2 92.67 147.34 <o,f0,male> well, Wei-sue, the situation clearly shaking investor sentiment there, there are worries that MSCI's move could trigger a flight capital, a flight of capital out of Indonesia, where the country's government policy, as you rightly pointed out, has previously, already generated fiscal risk concerns. Now some analysts have said that MSCI's announcement further highlights persistent concerns by foreign investors over ownership transparency. And now we also saw questions arise in recent days, as you mentioned earlier on Thomas appointment as deputy of Bank Indonesia, that's Indonesia's central bank. And that sparked debate on the independence of the institution as well. And remember, Mister Thomas's president, Brabo Su Bianto's nephew, and he was nominated for the post by the president as well. So given all of this, we could certainly, uh, see more volatility in Indonesia equities.
e-1770095240637 1 channel1_S1 150.85 163.19 <o,f0,male> um, this volatility is possible, but then again, this MSCI, this is a warning. And arguably, we could see improvements in those issues, as you mentioned, on transparency
e-1770095240637 1 channel1_S2 166.93 219.24 <o,f0,male> Well, at this point, we know that MSCI has said that if Indonesia has not made what it deems sufficient progress by the month of May, Indonesia could then be looking at a potential downgrade to frontier market status, which is below its, uh, current emerging market status. And MSCI will reassess market accessibility for global investors. Having said that, the Indonesia Stock Exchange said that it has been, it had been rather in discussions with MSCI regarding a free flow data for February 2026 rebalancing since the end of last year. And we've seen reports say that IDX claims the team made changes to the free flow of domestic shares. However, the process that they've done, it was deemed insufficient by MSCI, and now, IDX is working to resolve this, uh, all of these issues by the May 2026 deadline. So we'll suddenly be keeping a close watch to see what happens next.
e-1770095240637 1 channel1_S1 226.29 229.98 <o,f0,male> Um, thanks for that. CNA's Chanievani reporting live, does they're from Jakarta?
```

STM



Task 2 (Transcript downloads in multiple formats)

[00:00:00] channel1_S1: oil and business sharp declines for Indonesian shares, the benchmark plunging nearly 9% at one point, triggering a 30 minutes circuit breaker. The Jakarta Composite ending down more than 7% after MSCI warned about the market's instability. DNS provider wants regulators to address concerns that major companies are thinly traded and dominated by a few wealthy owners. It's warning of a possible downgrade if progress is not made by May. Indonesia's rupiah steady today. It has been under sustained pressure for months, tumbling to an all time low against the U.S. dollar.

[00:00:39] channel1_S1: currency has been sliding since August 2025 on concerns about fiscal discipline, or is deepening after the resignation of veteran finance Sri Mulliani Indrawati in September the same month. Uh, Bank Indonesia made two surprise rate cuts in line with the government's pro-growth push, raising concerns over its independence. Those concerns deepening in January with the nomination of President Proposal's nephew for Bank Indonesia's deputy governor with a CNA Channi to explain more about this, uh, stock market route

[00:01:13] channel1_S1: Uh, Shawnie.

[00:01:16] channel1_S1: concerns, confidence, both about central bank and about fiscal disciplines. In Indonesia, what will this mean this route for confidence? That's already at this point fragile.

[00:01:32] channel1_S2: well, Wei-sue, the situation clearly shaking investor sentiment there, there are worries that MSCI's move could trigger a flight capital, a flight of capital out of Indonesia, where the country's government policy, as you rightly pointed out, has previously, already generated fiscal risk concerns. Now some analysts have said that MSCI's announcement further highlights persistent concerns by foreign investors over ownership transparency. And now we also saw questions arise in recent days, as you mentioned earlier on Thomas appointment as deputy of Bank Indonesia, that's Indonesia's central bank. And that sparked debate on the independence of the institution as well. And remember, Mister Thomas's president, Brabo Su Bianto's nephew, and he was nominated for the post by the president as well. So given all of this, we could certainly, uh, see more volatility in Indonesia equities.

[00:02:30] channel1_S1: um, this volatility is possible, but then again, this MSCI, this is a warning. And arguably, we could see improvements in those issues, as you mentioned, on transparency

[00:02:46] channel1_S2: Well, at this point, we know that MSCI has said that if Indonesia has not made what it deems sufficient progress by the month of May, Indonesia could then be looking at a potential downgrade to frontier market status, which is below its, uh, current emerging market status. And MSCI will reassess market accessibility for global investors. Having said that, the Indonesia Stock Exchange said that it has been, it had been rather in discussions with MSCI regarding a free flow data for February 2026 rebalancing since the end of last year. And we've seen reports say that IDX claims the team made changes to the free flow of domestic shares. However, the process that they've done, it was deemed insufficient by MSCI, and now, IDX is working to resolve this, uh, all of these issues by the May 2026 deadline. So we'll suddenly be keeping a close watch to see what happens next.

[00:03:46] channel1_S1: Um, thanks for that. CNA's Chanievani reporting live, does they're from Jakarta?

TXT



Task 2 (Transcript downloads in multiple formats)

```
1
00:00:00,090 --> 00:00:34,630
oil and business sharp declines for Indonesian shares, the benchmark plunging nearly 9% at one point, triggering a 30 minutes circuit breaker. The Jakarta Composite ending down more than 7% after MSCI warned about the market's instability. DNS provider wants regulators to address concerns that major companies are thinly traded and dominated by a few wealthy owners. It's warning of a possible downgrade if progress is not made by May. Indonesia's rupiah steady today. It has been under sustained pressure for months, tumbling to an all time low against the U.S. dollar.

2
00:00:39,149 --> 00:01:10,460
currency has been sliding since August 2025 on concerns about fiscal discipline, or is deepening after the resignation of veteran finance Sri Mulliani Indrawati in September the same month. Uh, Bank Indonesia made two surprise rate cuts in line with the government's pro-growth push, raising concerns over its independence. Those concerns deepening in January with the nomination of President Proposal's nephew for Bank Indonesia's deputy governor with a CNA Channi to explain more about this, uh, stock market route

3
00:01:13,150 --> 00:01:14,269
Uh, Shawnie.

4
00:01:16,650 --> 00:01:28,629
concerns, confidence, both about central bank and about fiscal disciplines. In Indonesia, what will this mean this route for confidence? That's already at this point fragile.

5
00:01:32,670 --> 00:02:27,340
well, Wei-sue, the situation clearly shaking investor sentiment there, there are worries that MSCI's move could trigger a flight capital, a flight of capital out of Indonesia, where the country's government policy, as you rightly pointed out, has previously, already generated fiscal risk concerns. Now some analysts have said that MSCI's announcement further highlights persistent concerns by foreign investors over ownership transparency. And now we also saw questions arise in recent days, as you mentioned earlier on Thomas appointment as deputy of Bank Indonesia, that's Indonesia's central bank. And that sparked debate on the independence of the institution as well. And remember, Mister Thomas's president, Brabo Su Bianto's nephew, and he was nominated for the post by the president as well. So given all of this, we could certainly, uh, see more volatility in Indonesia equities.

6
00:02:30,849 --> 00:02:43,189
um, this volatility is possible, but then again, this MSCI, this is a warning. And arguably, we could see improvements in those issues, as you mentioned, on transparency

7
00:02:46,930 --> 00:03:39,240
Well, at this point, we know that MSCI has said that if Indonesia has not made what it deems sufficient progress by the month of May, Indonesia could then be looking at a potential downgrade to frontier market status, which is below its, uh, current emerging market status. And MSCI will reassess market accessibility for global investors. Having said that, the Indonesia Stock Exchange said that it has been, it had been rather in discussions with MSCI regarding a free flow data for February 2026 rebalancing since the end of last year. And we've seen reports say that IDX claims the team made changes to the free flow of domestic shares. However, the process that they've done, it was deemed insufficient by MSCI, and now, IDX is working to resolve this, uh, all of these issues by the May 2026 deadline. So we'll suddenly be keeping a close watch to see what happens next.
```

SRT



Task 2 (Transcript downloads in multiple formats)

[00:00:00] **channel1_S1:** oil and business sharp declines for Indonesian shares, the benchmark plunging nearly 9% at one point, triggering a 30 minutes circuit breaker. The Jakarta Composite ending down more than 7% after MSCI warned about the market's instability. DNS provider wants regulators to address concerns that major companies are thinly traded and dominated by a few wealthy owners. It's warning of a possible downgrade if progress is not made by May. Indonesia's rupiah steady today. It has been under sustained pressure for months, tumbling to an all time low against the U.S. dollar.

[00:00:39] **channel1_S1:** currency has been sliding since August 2025 on concerns about fiscal discipline, or is deepening after the resignation of veteran finance Sri Mulliani Indrawati in September the same month. Uh, Bank Indonesia made two surprise rate cuts in line with the government's pro-growth push, raising concerns over its independence. Those concerns deepening in January with the nomination of President Proposal's nephew for Bank Indonesia's deputy governor with a CNA Channi to explain more about this, uh, stock market route

[00:01:13] **channel1_S1:** Uh, Shawnie.

[00:01:16] **channel1_S1:** concerns, confidence, both about central bank and about fiscal disciplines. In Indonesia, what will this mean this route for confidence? That's already at this point fragile.

[00:01:32] **channel1_S2:** well, Wei-sue, the situation clearly shaking investor sentiment there, there are worries that MSCI's move could trigger a flight capital, a flight of capital out of Indonesia, where the country's government policy, as you rightly pointed out, has previously, already generated fiscal risk concerns. Now some analysts have said that MSCI's announcement further highlights persistent concerns by foreign investors over ownership transparency. And now we also saw questions arise in recent days, as you mentioned earlier on Thomas appointment as deputy of Bank Indonesia, that's Indonesia's central bank. And that sparked debate on the independence of the institution as well. And remember, Mister Thomas's president, Brabo Su Bianto's nephew, and he was nominated for the post by the president as well. So given all of this, we could certainly, uh, see more volatility in Indonesia equities.

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[00:02:46] **channel1_S2:** Well, at this point, we know that MSCI has said that if Indonesia has not made what it deems sufficient progress by the month of May, Indonesia could then be looking at a potential downgrade to frontier market status, which is below its, uh, current emerging market status. And MSCI will reassess market accessibility for global investors. Having said that, the Indonesia Stock Exchange said that it has been, it had been rather in discussions with MSCI regarding a free flow data for February 2026 rebalancing since the end of last year. And we've seen reports say that IDX claims the team made changes to the free flow of domestic shares. However, the process that they've done, it was deemed insufficient by MSCI, and now, IDX is working

Page 1 of 1 545 words  English (Singapore) Text Predictions: On  Focus    - 100%

DOCX



Task 2 (Transcript downloads in multiple formats)

```
File type = "ooTextFile"
Object class = "TextGrid"

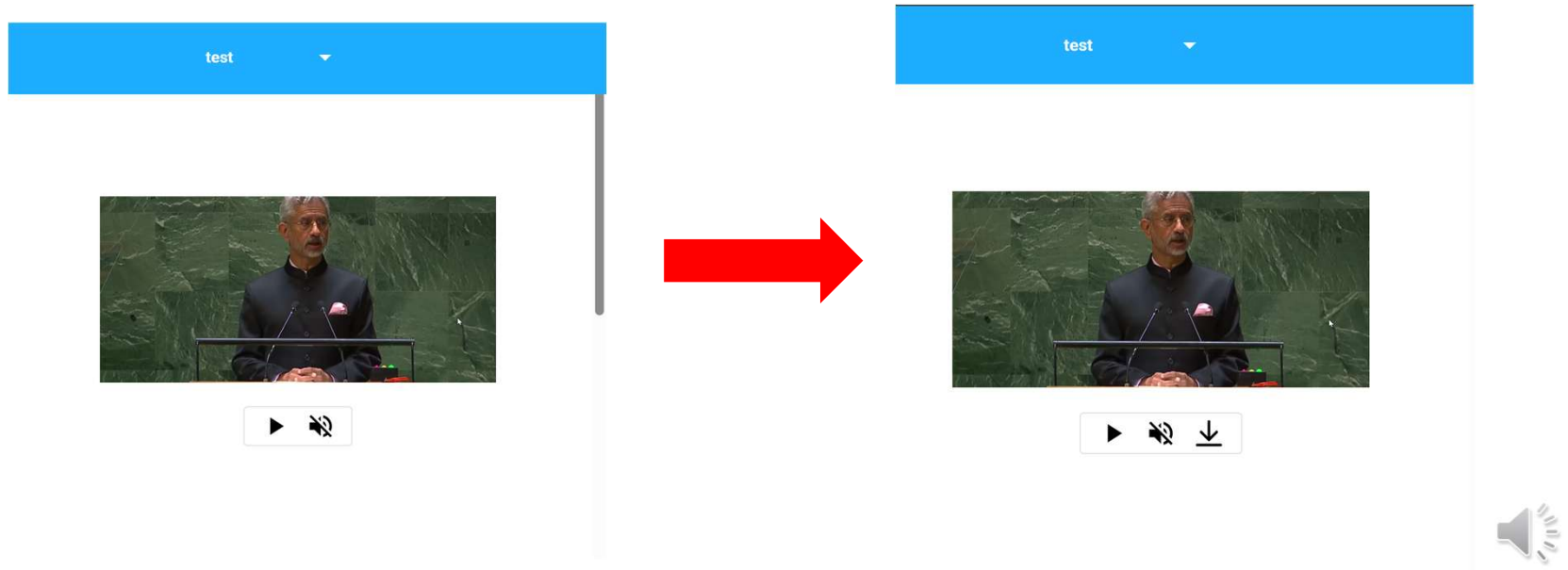
xmin = 0.09
xmax = 229.98
tiers? <exists>
size = 2
item []:
  item [1]:
    class = "IntervalTier"
    name = "channell_S1"
    xmin = 0.09
    xmax = 229.98
    intervals: size = 11
    intervals [1]:
      xmin = 0.09
      xmax = 34.63
      text = "oil and business sharp declines for Indonesian shares, the benchmark plunging nearly 9% at one point, triggering a 30 minutes circuit breaker. The Jakarta Composite ending down more than 7% after MSCI warned about the market's instability. DNS provider wants regulators to address concerns that major companies are thinly traded and dominated by a few wealthy owners. It's warning of a possible downgrade if progress is not made by May. Indonesia's rupiah steady today. It has been under sustained pressure for months, tumbling to an all time low against the U.S. dollar."
    intervals [2]:
      xmin = 34.63
      xmax = 39.15
      text = ""
    intervals [3]:
      xmin = 39.15
      xmax = 70.46000000000001
      text = "currency has been sliding since August 2025 on concerns about fiscal discipline, or is deepening after the resignation of veteran finance Sri Mulliani Indrawati in September the same month. Uh, Bank Indonesia made two surprise rate cuts in line with the government's pro-growth push, raising concerns over its independence. Those concerns deepening in January with the nomination of President Proposal's nephew for Bank Indonesia's deputy governor with a CNA Channi to explain more about this, uh, stock market route"
    intervals [4]:
      xmin = 70.46000000000001
      xmax = 73.15
      text = ""
    intervals [5]:
      xmin = 73.15
      xmax = 74.27
      text = "Uh, Shawnie."
    intervals [6]:
      xmin = 74.27
      xmax = 76.65
      text = ""
```

TEXTGRID



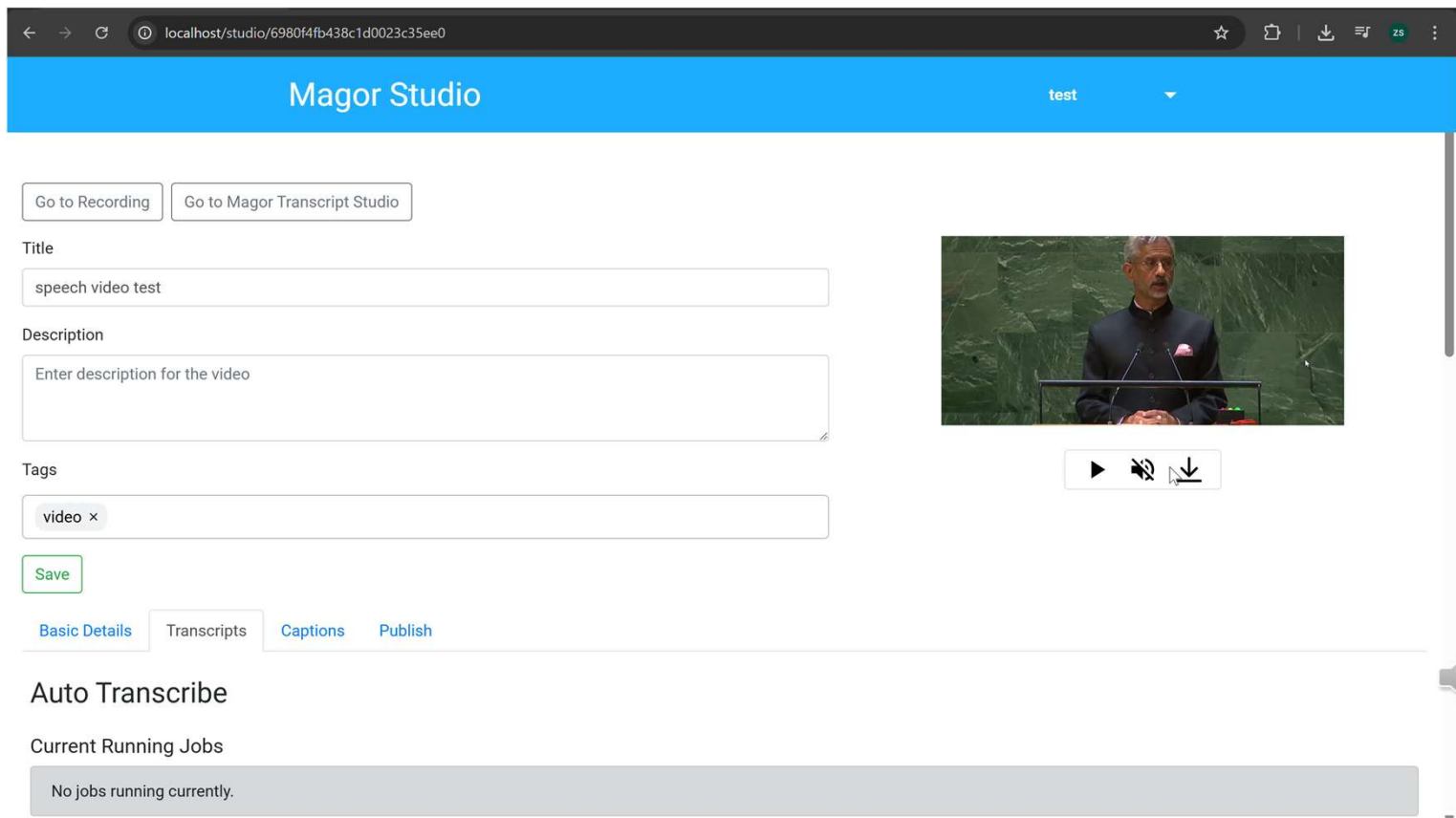
Task 3 (Media download on the Studio page)

- Implemented button to download uploaded media on the Studio Page



Task 3 (Media download on the Studio page)

- Demo:



Task 4 (UI refinements)

- Moved pagination below recording list
- Adjusted alignments of columns (**WIP**)

The diagram illustrates the UI refinements for a recording list. On the left, the current state shows a table with columns: Title, Created, and Recording Type. The pagination is located above the table, showing "Showing 1 to 10 of 4 result(s)". A red box highlights the pagination and the column headers. A red arrow points from this box to the refined version on the right. The refined version shows the pagination moved below the table, showing "Showing 1 to 3 of 3 result(s)". The column headers are also refined, with "Created" and "Recording Type" aligned to the right. The table content is also refined, with the "Recording Type" column now containing "View", "Edit", and "Delete" buttons.

Title	Created	Recording Type
Indonesian stocks plunge more than 7 after MSCI flags risks - CNA (youtube).mp3	an hour ago	View Edit Delete
audio 2 test	a day ago	View Edit Delete
Original_recording5.mp4	2 days ago	View Edit Delete
harvard.wav	2 days ago	View Edit Delete

< Previous Page Showing 1 to 10 of 4 result(s) Next Page >

Magor test

Magor tes

Title	Created	Recording Type
Indonesian stocks plunges	2 days ago	View Edit Delete
harvard audio	2 days ago	View Edit Delete
speech video test	2 days ago	View Edit Delete

< Previous Page Showing 1 to 3 of 3 result(s) Next Page >



Plans for Next Week

- Fix Textgrid conversion
 - Replace empty string "" with "--EMPTY--"
- Fix Srt conversion
 - add speaker id
- Check if edited transcripts are reflecting changes when downloaded
- Re-align the columns of the uploaded recordings list
- Further UI refinements and bug fixes
 - Top bar direct navigation to pages

